Observation Robustness in Bimanual Handover — v3

Sensor bias, noise and delay vs geometry and residual PPO · Isaac Lab 3.0 / Isaac Sim 6.2 (PhysX) · CSCI 645, Fall 2026 · Yan Wen

- Two Franka arms, parallel-jaw grippers, one rigid bar; success = 2 s receiver-only hold
- v3 corrupts only the bar's translation measurement (synthetic sensor, not vision)
- Held-out: 135,200 logical episodes (6 baselines + 23 PPO runs × selected/final × 2,600 cases)
- Strong estimator saturates the suite: geometry_common 100.0% on mixed_id
- Both primary contrasts are exactly 0 pp: a ceiling, not a win for learning
- Release v3.0.0: every checkpoint, raw record, log, video and checksum

Where v2 left off

previous studies, quoted from the v2 report (release v2.1.1)

v2 controller (accurate state)	all-case success
nominal script	65.2%
residual PPO	82.6%
geometry	96.5%
combined (geometry + residual PPO)	96.1%
geometry + tuned constant	99.2%

- v2: learned residuals on top of geometry ≈ a small constant offset
- Open question: does geometry break when the object estimate is corrupted?
- Hypotheses (frozen before held-out data):
  - H1 bias / noise / delay hurt controllers differently
  - H2 noise-trained residuals beat clean-trained ones
  - H3 a 5-frame history helps beyond input size
  - H4 filtering + giver-grasp reference remove that benefit

Synthetic translation sensor

- $y_k = x_{k-d} + b + \sigma \cdot n(\text{seed}, k)$; estimate $\hat{y}_k = \alpha \cdot y_k + (1-\alpha) \cdot \hat{y}_{k-1}$
- Same noise sequence for every controller on a case (counter-based generator)
- Giver reference: while the giver holds, estimate = giver TCP − nominal grasp offset (proprioception)
- Orientation, joints, TCPs, gripper widths and contact forces stay accurate
- Final suite: 200 base cases × 13 conditions = 2,600 paired cases
  - clean, physical_only, bias 2/5/10 mm, noise 1/2/4 mm, delay 1/2/4 steps (50 ms each)
  - mixed_id: bias ≤ 5 mm, σ 2 mm, delay 2 · mixed_ood: bias ≤ 10 mm, σ 4 mm, delay 4
- Validation 240 cases and development 312 cases on separate seeds

A controlled model of translation-estimation error — not a vision system and not a sim-to-real claim.

Controllers and learned policies

controller	estimator	learned
nominal	none (fixed schedule)	–
geometry_raw	α = 1, no giver reference	–
geometry_filtered	EMA α = 0.2	–
geometry_common (= best_constant = giver)	α = 1 + giver reference	–
combined_clean_ff / robust_ff	geometry_common	PPO residual, 1 frame (46 inputs)
combined_robust_history	geometry_common	PPO residual, 5 frames (230 inputs)
combined_robust_repeat	geometry_common	5 × current frame (memory control)
residual_robust_history	on nominal script	PPO residual, 5 frames

- Residual = 0.02·tanh(a) m per axis; PPO (RSL-RL 5.5.1), MLP [128,128]; 10,027,008 transitions per run
- Seeds 11/22/33/44/55 (repeat control: 11/22/33) → 23 formal runs

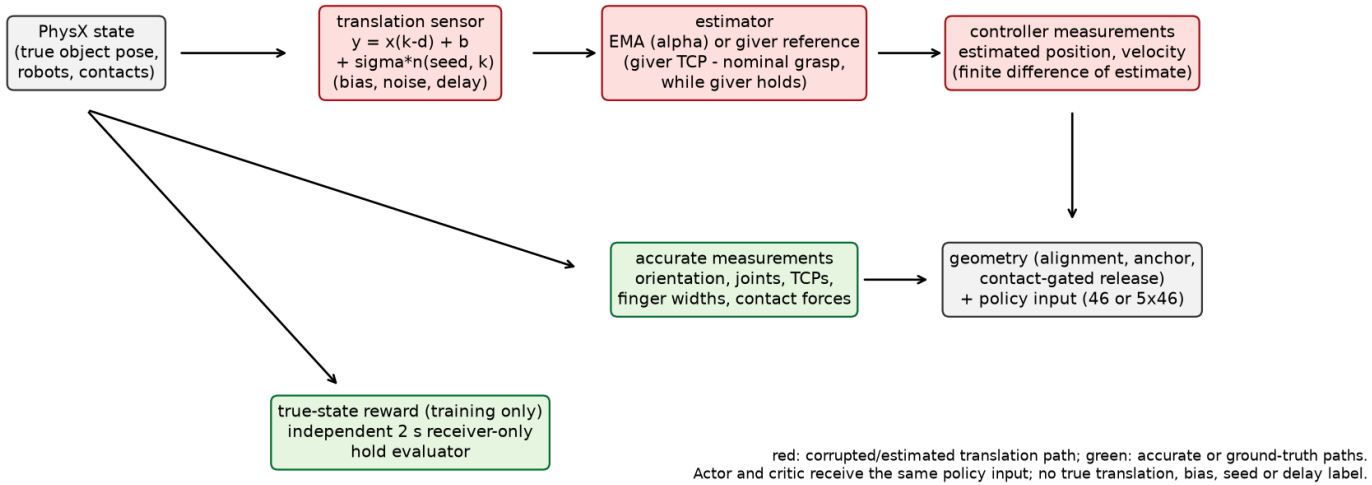
Protocol: develop → freeze → train → validate → test

- Remote acceptance: 10 of 10 checks passed (sensor, resets, no hidden truth path, PPO resume, caches)
- Development search over estimators and constants on development cases only
- Pilots ≤ 24M transitions → PPO settings unchanged; protocol frozen as tag experiment-v3
- 23 runs × 10M transitions: 7.6 A100 GPU-hours
- Selection on 240 validation cases at 2/4/6/8/10M (step 0 = control, never eligible)
- Final: one fresh process per entry, batch 256, all 2,600 cases; 0 invalid resets
- Primary (mixed_id, selected): history − robust FF; history − geometry_best_constant
- Paired bootstrap, 10,000 replicates over seeds and base-case clusters; 97.5% Bonferroni

Development: the estimator decides

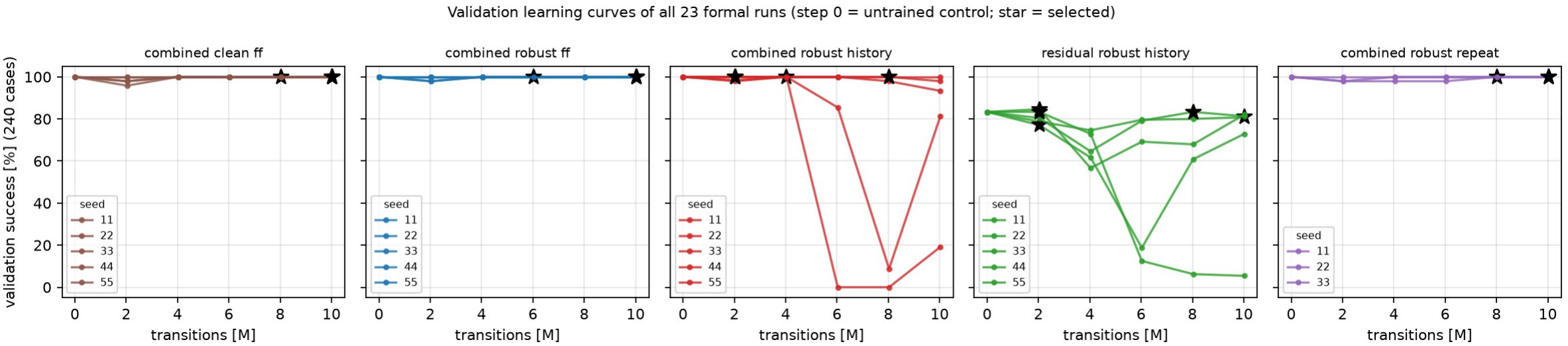
development cases, used only to choose baselines

- Raw estimate + noise ≥ 2 mm: release gate never opens (timeouts)
- EMA filtering helps noise; giver reference fixes it for every smoothing level
- Delay up to 4 steps: no effect (the bar is nearly stationary before release)
- Best constant offset = 0 → geometry_best_constant = geometry_common
- This was decided before any validation or final data



red: corrupted/estimated translation path; green: accurate or ground-truth paths.
Actor and critic receive the same policy input; no true translation, bias, seed or delay label.

Validation learning curves (all 23 runs)



- Combined runs at 100% from step 0; selection fell to the residual-size / latest-step tie-breaks
- History training is unstable: seed 55 0.0% at 6M, seed 11 8.8% at 8M
- Residual-only history declines after 2–4M (seed 55: 5.4% at 10M)

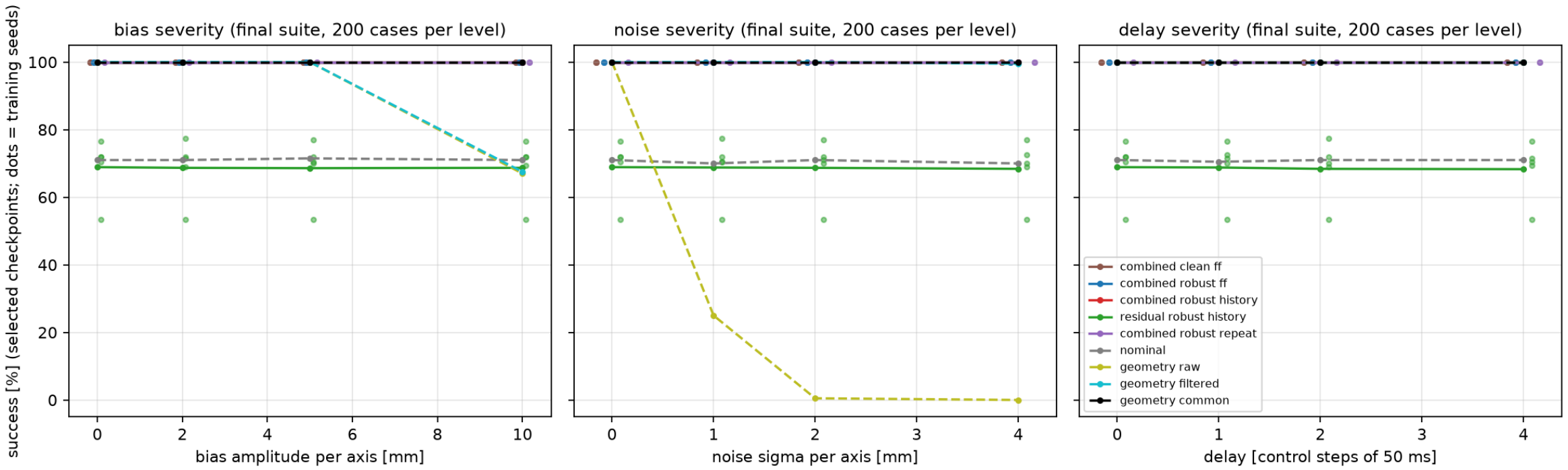
Held-out success (%) by condition

selected checkpoints; learned rows = mean over seeds; 200 cases per cell

controller	clean	physical only	bias 10mm	noise 4mm	delay 4step	mixed id	mixed ood
nominal	100.0	71.0	71.0	70.0	71.0	70.5	71.0
geometry_raw	100.0	100.0	67.0	0.0	100.0	0.0	0.0
geometry_filtered	100.0	100.0	67.5	99.5	100.0	100.0	96.0
geometry_common	100.0	100.0	100.0	100.0	100.0	100.0	100.0
combined clean FF	100.0	100.0	100.0	100.0	100.0	100.0	100.0
combined robust FF	100.0	100.0	100.0	100.0	100.0	100.0	100.0
combined robust history	100.0	100.0	100.0	100.0	100.0	100.0	100.0
combined repeated-current	100.0	100.0	100.0	100.0	100.0	100.0	100.0
residual-only history	100.0	68.9	68.7	68.4	68.3	68.7	68.8

Full 13-condition table with final checkpoints: report Section 6.2.

Severity curves (selected checkpoints, per-seed dots)



- Noise: raw geometry 25.0% at 1 mm → 0.5% at 2 mm (release timeouts)
- Bias 10 mm: raw 67.0%, filtered 67.5% (collisions); giver reference 100%

Primary contrasts: exactly zero

mixed_id, selected checkpoints, 5 seeds × 200 paired cases

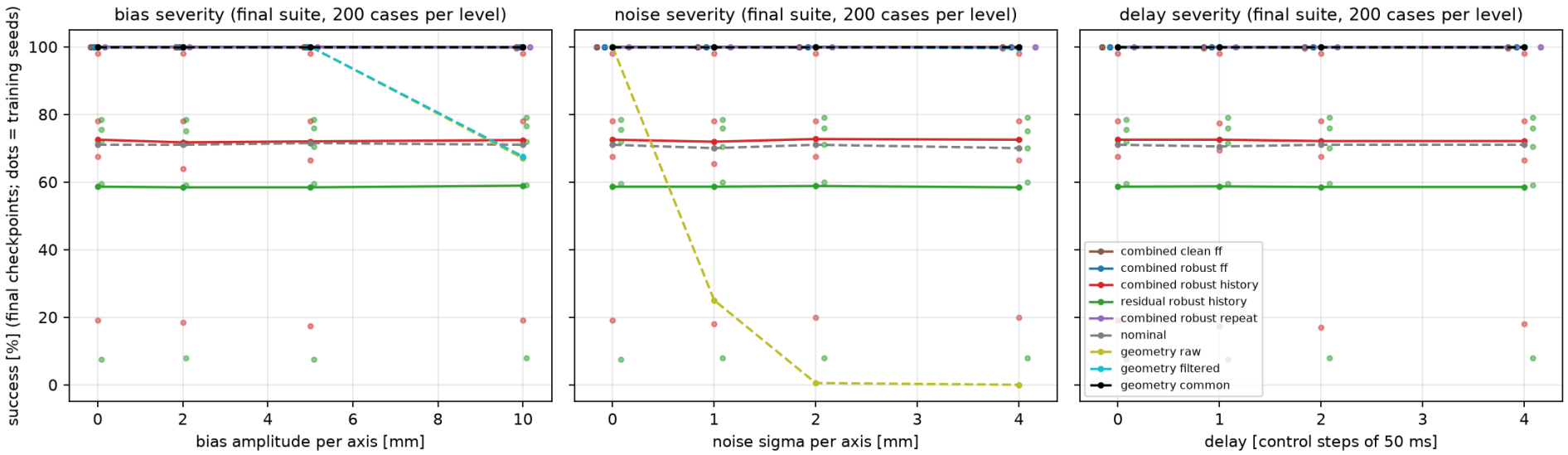
contrast	difference [pp] (95% CI)	97.5% Bonferroni	per seed
history − robust FF	+0.0 [+0.0, +0.0]	[+0.0, +0.0]	+0.0 / +0.0 / +0.0 / +0.0 / +0.0
history − geometry_best_constant	+0.0 [+0.0, +0.0]	[+0.0, +0.0]	+0.0 / +0.0 / +0.0 / +0.0 / +0.0
robust FF − clean FF (secondary)	+0.0 [+0.0, +0.0]	–	+0.0 / +0.0 / +0.0 / +0.0 / +0.0
history − repeated-current (secondary)	+0.0 [+0.0, +0.0]	–	+0.0 / +0.0 / +0.0
residual-only history − nominal (secondary)	−1.8 [−10.2, +4.4]	–	+0.0 / −17.0 / +6.5 / +1.5 / +0.0

- Zero-width intervals: both sides succeed on every case — a ceiling, not proof of equivalence
- No evidence that history or noise-trained residuals help on this suite (H2, H3 untestable here)
- Residual-only history does not beat the nominal script (unlike v2 with accurate state)

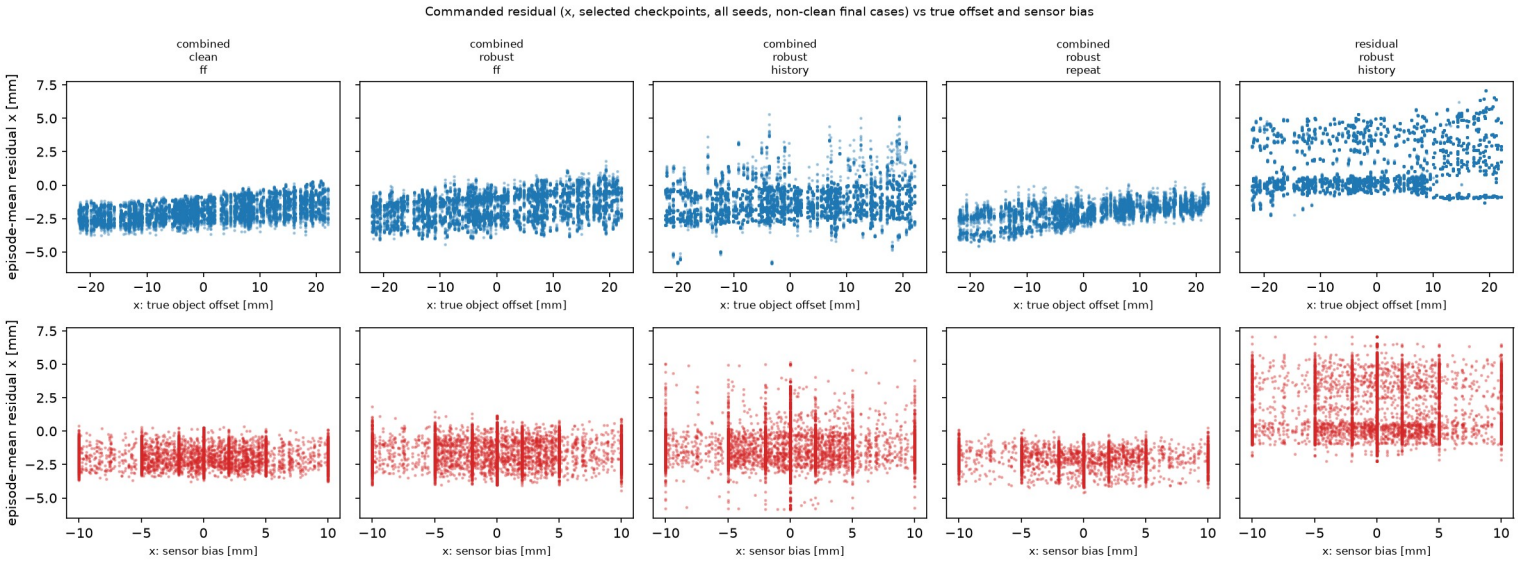
Final (10M) checkpoints: selection mattered

declared secondary analysis

- History final − geometry_common, mixed_id: −28.2 [−56.4, −5.4] pp
  - per seed −35.5 / +0.0 / −22.0 / −2.0 / −81.5
- Validation at 10M predicted it: seed 55 19.2% (collisions)
- Feed-forward and repeated-current final checkpoints stay at ceiling
- Residual-only history final − nominal: −12.1 [−38.4, +5.7] pp



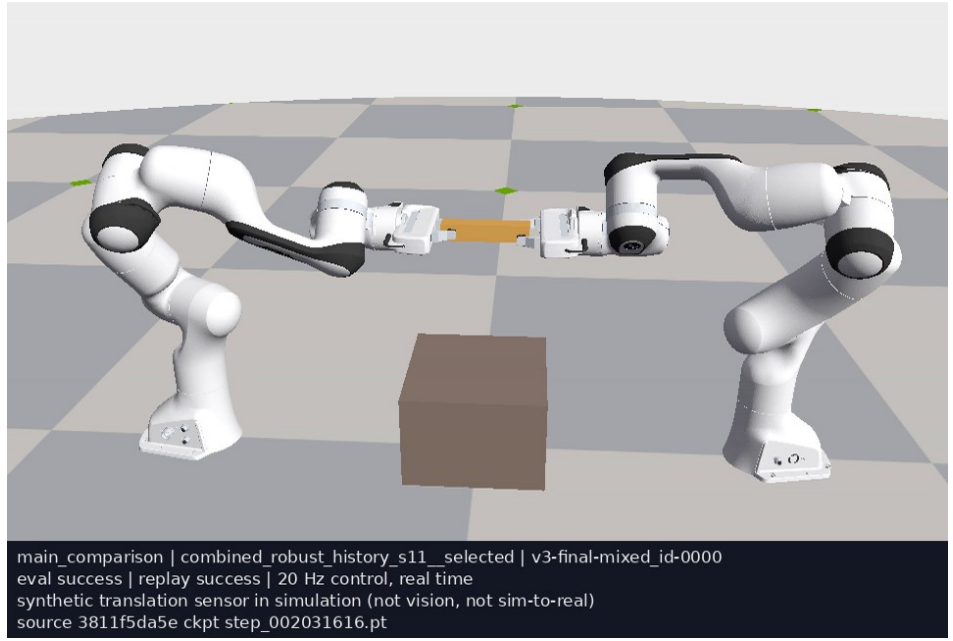
What the residuals do



- Combined: episode-mean x offset ≈ constant, -2.6 to -0.4 mm
- Slope on injected bias ≤ 0.008: ignores the sensor error
- Mean |residual| 2.0–7.8 mm, max 17.2 of 20 mm; no clipping
- Frames permuted / current-only: 325 and 325 of 325 successes (unmodified 325)
- Step-0 policies reproduce their controller: 65/65

Repeatability and videos

- 65-case subset × 7 entries × 3 process seeds × batch 1/23: 2,730 of 2,730 terminal outcomes reproduced
- History − geometry_common: +0.0 [+0.0, +0.0] pp (batch 1), +0.0 [+0.0, +0.0] pp (batch 23)
- Sensor parameters identical in every rerun; a check at the ceiling, not of a nonzero effect
- Videos: 8 clips by a frozen rule + 43 s montage; 20 Hz, real time
  - No main-comparison failure exists in mixed_id / mixed_ood → none shown, none substituted
  - Evaluation vs single-env replay disagreements: 0



Conclusions

- H1 supported: noise breaks the raw estimate at the release gate; 10 mm bias causes collisions; delay is harmless
- H4 supported: a giver-grasp reference makes geometry perfect on all 2,600 held-out cases
- H2, H3 not testable: every selected combined policy is also at 100%
- History training was the least stable; validation selection prevented a large loss
- Learned residuals act as small constant offsets and ignore the sensor error
- Next step (would be a new study): harder suites, e.g. error after release, object motion, grasp slip

Limitations and artifacts

- Ceiling makes the primary contrasts uninformative; one object, fixed physics
- Giver reference assumes a rigid grasp with known offset and accurate giver proprioception
- Synthetic translation sensor only; simulation only; 5 seeds (3 for the memory control)
- Physical-transfer extension not performed (spawn overrides not validated before freeze)
- Release v3.0.0: source (experiment-v3 + v3.0.0), all checkpoints, raw records, logs, videos,
  - release_manifest.json, SHA256SUMS, archive inventories; fresh-download verification report
- AI-assisted: implemented, run and written by an AI coding agent (Claude Code) for the owner